lúbadh

_travelers guide _for navigating time loops

2	10	4
decks · stereo link	min per loop, per deck	taps per deck

[rec] press [rec] hold [shift] = retrig/shift opp. = the other deck's button › then
◆ = preset / mode only

every control is per deck: left and right are identical

combos shown for one deck · mirror them for the other

amber = hold

01 transport — lbh_01

[rec]	record › press again to stop & play. press while playing = **overdub**
[erase]	erase this deck's loop
[shift]	jump back to start
turn [speed]	speed & direction: center = stall, cw = forward to 4×, ccw = reverse to 4×
[shift] + turn [speed]	knob is muted until release, then jumps straight to the new speed
[link] up	right deck mirrors left · saves and loads stereo
[shift]	adds a playhead at start (up to 4)
◆ multi-tap presets	

02 recording — lbh_02

[erase] + [rec]	**punch-in**: replaces audio while recording
[shift] [shift] + [rec]	**input monitoring**: on (rec lit) › armed (rec pulses) › off
armed [rec]	arm › record › stop & play · [erase] disarms, loop kept
[shift] + opp. [rec]	looping ← → **one-shot record** (records one pass, then stops)
[rec] jack	trig starts / stops recording, like [rec] (latching)
[rec] jack	gated: records only while the gate is high
◆ rec jack = gated	

* overdub keeps the old layer at 0.9 by default, so layers fade a little each pass (dub level, 04).

__fig.01 : speed

white speed LEDs = notches



4× rev	½ rev	stall	½	1×	2×	4×
+24 st	-12 st	–	-12 st	0 st	+12 st	+24 st

speed sets pitch: every doubling = +1 octave · reverse plays backwards

03 playback — lbh_03

[shift] + opp. [erase]	looping ← → **one-shot playback** (rec blinks white / mixed)
[shift] or [retrig] jack	one-shot: play one pass of start › length
touch the reel	warble or stall the tape · trimmer sets sensitivity
[speed] cv	±5 v, summed with the knob
[erase] + opp. [erase]	flip playback direction · speed LEDs blink
◆ v/oct speed mode only	
[speed] cv	tracks 1v/oct: sequence the loop's pitch
◆ v/oct · seq presets	

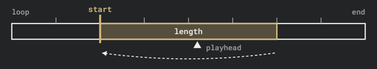
04 time knob — lbh_04

[shift] + [rec]	cycle the time function (amber LED by [clk out] shows which)
LED off	**clock divisions**: triggers per loop at [clk out]: 1–12, 16, 24, 32, 64
LED solid	**dub level**: how much old audio survives each overdub. high = delay feedback
LED pulsing	**user parameter**: crossfade, tape or slew (preset picks)



05 start · length — lbh_05

turn [start]	where the loop begins inside the recording
turn [length]	how much plays before jumping back. tiny = wavetable-like tones
[shift] + turn [time]	**quantise** start & length: lit LED sections = 1/2, 1/4, 1/8 …
[shift] + turn [start]	fine move that ignores the quantise grid (same for [length])



ticks = quantise grid (1/8)

06 save · load — lbh_06

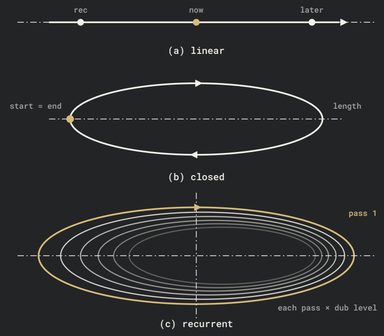
[shift] [erase] + [rec]	**save** to usb › saved (folder 0, newest first)
[shift] + [erase]	**load menu**
in the menu	own [time] = loop · opposite [time] = folder · [rec] loads · hold [shift] previews 5 s · [erase] exits
[link] up	saves and loads stereo (mono loads of stereo are summed)
power-on	last session reloads automatically

07 presets — lbh_07 · v2.1

[shift] [shift] + [erase]	**preset menu** for that deck
turn [time]	pick 1 of 12 (shown on the LED ring) · [rec] loads · [erase] cancels
usb › presets	.txt files, sorted 01–12. build your own with the preset editor

__fig.02 : the return

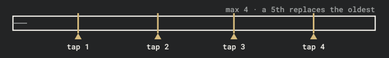
mm_02



08 multi-tap — lbh_08

◆ multi-tap presets only

[shift]	new playhead from start
[erase] + [shift]	delete the taps
[erase] trig, cv exp.	removes taps remotely



09 sync · clock — lbh_09

[clk out]	triggers per loop (set by clock div)
[retrig] jack	resets the playhead to start: sync to an external clock
[rec] jack	external clock sets the loop length
◆ clockable preset	
[erase] + [rec]	tap tempo instead of punch-in
◆ erase + rec = tap tempo	

10 aux · routing — lbh_10

aux in fader	sends [aux] in to the left ← → right deck input (+ [cv] ±5 v)
aux out fader	blends left ← → right outputs at [aux] out (+ [cv] ±5 v)
rear switch	aux out fader pre or post the output level, per deck



11 reset · calibrate — lbh_11

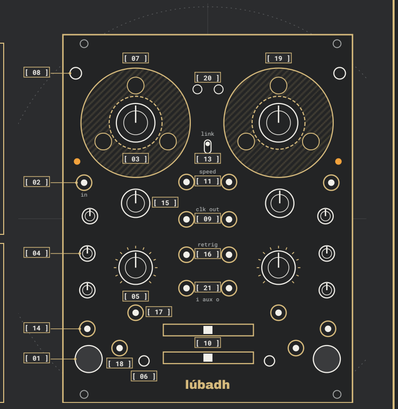
[shift] [shift] + [rec] [rec]	**reset modes** to default · audio kept · speed LEDs pulse
power on, hold [shift]×2 [erase]×2	**calibration**: speed knobs min, 0 v › [rec] each deck, then 1 v and 3 v; knobs max, no cv › [rec] = 5 v
amber blinks	voltage out of range: defaults used

__fig.03 : front panel

01–11 = sections · 13–21 = hardware

left deck labelled · right deck mirrors it

[01] __rec : record · overdub · stop	[13] __link : stereo mirror
[02] __in : input + level	[14] __out : output + level
[03] __speed : rate & direction	[15] __length : loop length
[04] __time : clock div · dub · user	[16] __retrig : jack, restart
[05] __start : loop start	[17] __cv : aux fader cv
[06] __erase : erase · save / load	[18] __rec : jack, record gate
[07] __LED ring : position · presets	[19] __reel : capacitive touch
[08] __retrig/shift : restart · shift	[20] __speed LEDs : notches
[09] __clk out : clock triggers	[21] __aux : in · out jacks
[10] __aux faders : in / out crossfade	
[11] __speed cv : ±5 v · 1v/oct	



_where things hide

12 · lbh_12

preset files

usb › presets › .txt · instruo preset editor

- speed record variable · fixed, play follow · hold
- speed control notched · stepped · smooth · v/oct, markers 0–4
- multi-tap engine off · on
- time user crossfade · speed slew · tape emulation
- rec jack latching · gated · external clock
- ext clock average, resolution, timeout, divisions
- quantise all · even · odd · powers of two
- erase + rec punch-in · tap tempo
- timing min loop, crossfade, retrig delay, max dub
- tape wow / flutter, crinkle, age, hysteresis, wear, saturation, reverb
- input filter low cut freq + q, high cut
- touch off · stall · dip

usb disk

- samples 12 folders × 12 loops
- saved folder 0, newest first
- presets first 12 show in the menu

tails & tags

250 ms, in the file name

- {s} silent tail (default): keeps timing
- {n} no tail: smoothest, timing loosens
- {c} copy tail: keeps timing, may click

hardware

- cv expander [length] [start] [time] cv + [erase] trig, per deck
- rear pcb touch on / off · pre / post output level
- front touch sensitivity trimmer

factory presets

- 01–05 tape looper · clean · classic · multi-tape · clockable
- 06–10 octave delay · seq mono · seq poly · melontron · broken tape

tails at the loop point

250 ms · gold = the crossfade

